Gebze Technical University FALL 2023 ELEC334 Microprocessors by Dr. Ihsan Cicek

Homework 2 Assigned: 21/11/2022 Due: 28/11/2022 17:00

1. (20 pts) Write an ARM Cortex M0+ assembly code which converts the data 0xC0DEBAD1 stored in R0 into 0x600DC0DE using Boolean and shift-rotate instructions. Show how your code executes step-by-step using line comments.

2. (20 pts) Write an ARM Cortex M0+ assembly code that swaps the data contents of R0=0x600DC0DE with R1=0x1BADC0DE without using an additional temporary storage register.

3. (20 pts) Write an ARM Cortex M0+ assembly code that calculates the multiplication of R0 content by q constant decimal number 0x0334 without using the MUL instruction. Execute your code for R0=0xE1EC and store the result in R0.

4. (20 pts) Write an assembly program to subtract 128-bit numbers from each other, placing the result in registers R3, R2, R1 and R0 in little-endian format. The first operand is placed in registers R7, R6, R5 and R4, and the second operand is in registers R11, R10, R9 and R8 in little-endian format.

5. (20 pts) Write an assembly program to add 128-bit numbers, placing the result in registers R3, R2, R1 and R0 in little-endian format. The first operand is placed in registers R7, R6, R5 and R4, and the second operand is in registers R11, R10, R9 and R8 in little-endian format.